INT104 Experiment Report

Yifei Li 2364053

Abstract

This experimental report first conducts feature processing of standardization, normalization (Min-Max scaling) and unit normalization on the original data, and then comprehensively analyzes it by using both unsupervised learning and supervised learning methods. In the unsupervised learning part, cluster analysis is conducted using K-Means, Gaussian Mixture Model (GMM), and hierarchical clustering algorithms, and the optimal feature collection and clustering methods are determined through the distance_ratio function. In the supervised learning part, the classifier is constructed by adjusting parameters such as the n-nearest neighbors of the k-nearest neighbors, the criteria of the decision tree, and the prior of Gaussian Naive Bayes. Evaluate the accuracy rates under different Settings and find the best classifier and feature combination. The experimental results provide a basis for an in-depth understanding of the relationship between student data characteristics and project classification, and also offer a reference for subsequent educational data analysis.

1. Introduction

Various information of the students was recorded in the samples of this experiment. The first column is the student ID. Gender codes are "1" and "2", grades are divided into "2" and "3", and projects are represented by "1" to "4" to indicate different professional directions or study plans. The dataset also includes the total score and the scores of five test questions (Q1-Q5), with full marks of 8, 8, 14, 10 and 6 respectively, which can reflect students' mastery of different knowledge points.

The experiment first conducts feature processing of standardization, normalization(Min-Max scaling) and unit normalization on the original data.

The subsequent experiments adopted unsupervised learning and supervised learning strategies.

(1) In unsupervised learning, data processing adopts three algorithms, namely K-Means, Gaussian Mixture Model (GMM), and hierarchical clustering, as well as multiple configurations. K-Means iteratively optimizes the clustering centers, GMM captures complex distributions, and hierarchical clustering builds a tree structure to determine the optimal feature collection and clustering methods.

(2) In the supervised learning part, this classifier is constructed by adjusting the n_neighbors parameter of the k-nearest neighbors, the criterion of the decision tree, and the prior parameters of Gaussian Naive Bayes. k nearest neighbors are classified by distance, and n nearest neighbors affect the local sensitivity. The decision tree is divided according to the features, a tree-like structure is constructed, and the division criteria are determined according to the criteria. Gaussian naive Bayes classification is based on Bayes' theorem and probability distribution classification, and the prior reflects prior cognition. Evaluate the accuracy of different classifiers, identify the best classifier and feature combination, and use it to predict the project to which the student belongs.

2. Data analysis and processing

The sample dataset contains approximately 300 records, covering 4 programmes, 2 genders, 2 grades, and 5 examination results (Q1-Q5, with full marks of 8, 8, 14, 10, and 6 respectively, and the score ranges vary significantly. The highest score in Q3 was 14 and that in Q5 was 6. Among them, the sample size of item "3" was relatively large while that of "4" was relatively small, indicating an imbalance in categories. Moreover, the distribution of scores was skewed, with some students' low scores concentrated in Q2 and Q5, and high scores concentrated in Q3 and Q4.

However, there are shortcomings such as dimensional differences and uneven distribution in the original data. Therefore, the following three conversion methods are adopted: standardization, normalization (Min-Max scaling), and unit normalization. Thereby optimizing the data distribution, adapting to the requirements of different algorithms, and improving the model performance.

(1)<u>Standardization</u>

$$X_{std} = \frac{X - \mu}{\sigma}$$

The feature is scaled to a standard normal distribution with a mean of 0 and a standard deviation of 1 through this formula

Advantages:

1.Eliminating dimensional influence: The full marks of Q1 to Q5 in the original data vary greatly (for example, the full

score of Q3 is 14 and that of Q5 is 6). After standardization, all features are on the same scale to avoid the model's bias towards high-score features (such as Q3).

2.Adaptive sensitive algorithms: Applicable to algorithms such as KNN, Support Vector Machine (SVM), and neural networks that are sensitive to feature scales. For example, KNN is classified based on Euclidean distance, and the distance calculation is fairer after standardization.

3.Retain distribution shape: It does not change the form of data distribution, only adjusting the position and scale. It is suitable for scenarios where the characteristics of data distribution need to be retained

(2)Normalization (Min-Max scaling)

$$X_{norm} = \frac{x - x_{min}}{x_{max} - x_{min}}$$

Scale the features to the interval of [0,1], retain the original distribution shape, and it is suitable for algorithms such as neural networks that require a fixed range.

Advantages:

1.Retain relative proportions: Maintain the relative size relationship among features, which is suitable for scenarios where the original distribution proportion of data needs to be retained (such as data visualization and probabilistic model input).

2.Simplified calculation: Limit the eigenvalues within a fixed interval to facilitate the processing of activation functions for models such as neural networks (for example, the output range of the Sigmoid function is [0, 1]).

3.Handling sparse data: More robust to extreme values of high segments (such as Q3=14) and low segments (such as Q5=0), avoiding numerical instability caused by too small a standard deviation after standardization.

(3)unit normalization

$$X_{unit} = \frac{X}{\|X\|_2}$$

Scaling the sample vector to the unit norm, emphasizing the direction rather than the modulus, is suitable for probabilistic models (such as naive Bayes).

Advantages:

1.Focus on directional features: It is applicable to scenarios such as text mining and image features that require attention to the similarity between samples (such as cosine similarity), for example, calculating the directional differences of student grade vectors.

2.Eliminating the influence of scale: Students with high total scores in the original data may dominate the distance calculation due to the larger modulus. After unit normalization, only the direction information is retained, which is suitable for comparing the score structure (such as the proportion of scores for each question type).

3.Adaptive probability models: Models such as Gaussian and Naive Bayes assume that features follow a normal distribution. Unit normalization can make the feature distribution closer to the standard normal and improve the accuracy of probability estimation.

The clustering index distance_ratio is used for feature comparison. The smaller the ratio, the more compact the intra-class and the better the separation between classes. K-Means clustering (n_clusters=4) was performed on the features after the three transformations respectively, and the distance_ratio was calculated. standardization = 0.42. After eliminating the dimension, the weights of each feature are consistent, and the intra-class distance is significantly reduced (such as the influence equilibrium of Q3 and Q5). normalization (Min-Max scaling) = 0.35, retaining the relative proportion. The high-score advantage of Q3 and the low-score disadvantage of Q5 are scaled to the same interval, making the clustering more compact. unit normalization = 0.49, only the direction information is retained, the absolute value differences (such as the total score differences) are ignored, and the inter-class separation degree is relatively low. Therefore, normalization (Min-Max scaling) has the best effect, retaining the relative proportion of the scores (such as the proportional relationship between 14 points in Q3 and 6 points in Q5), and avoiding dimensional interference through fixed intervals. It is suitable for distance dependent algorithms such as K-Means. standardization comes second, suitable for scenarios that are sensitive to scale but need to retain the shape of the distribution. unit normalization is limited by overemphasizing the direction, resulting in the loss of absolute value information and a poor clustering effect. It is not very suitable for the analysis of absolute value differences in performance data.

3. Unsupervised learning

In unsupervised learning, three methods, namely K-Means, GMM and hierarchical clustering, are used for clustering three features.

Configuration parameters	standardization	normalization (Min-Max scaling)	unit normalization
K-Means	n_init=10, init='k-means++', algorithm='elkan'	n_clusters=4, tol=1e-5, max_iter=500	n_clusters=4, precompute_distances=True, n_jobs=-1
GMM	n_components=4, covariance_type='spherical', reg_covar=1e-6	n_components=4, covariance_type='diag', init_params='kmeans'	n_components=4, covariance_type='full', warm_start=True
HC	method='ward', metric='euclidean', optimal_ordering=True	method='complete', metric='cosine', criterion='maxclust'	method='average', metric='cosine', criterion='inconsistent'

Reasons for experimental design

(1)K-Means:

1.standardization: The standardized data conforms to the assumption of data distribution by K-Means, and the elkan algorithm accelerates the calculation

2.normalization (Min-Max scaling): After normalization, the data range is consistent, but it may lead to dense clusters, and the convergence accuracy needs to be improved.

3.unit normalization: The pre-computed distance matrix accelerates the calculation between unit vectors, and multi-core parallelism improves efficiency.

(2)GMM:

1.standardization: Spherical assumes the isotropic distribution after adaptation standardization, and regularization prevents the covariance matrix from being singular.

2.normalization (Min-Max scaling): Diag assumes that features are independent of each other, and kmeans initialization enhances stability.

3.unit normalization: The full covariance matrix captures the complex relationships among unit vectors and accelerates the iteration through hot start.

(3)Hierarchical Clustering:

1.standardization: Ward minimizes variance, the Euclidean distance is suitable for standardized data, and the optimal sorting enhances the visualization effect.

2.normalization (Min-Max scaling): The complete link prevents large clusters from being dominant, and cosine similarity focuses on direction rather than value size.

3.unit normalization: The average link balances the cluster size, and the cosine similarity is suitable for the comparison of unit vector directions.

In contrast, standardization uses K-Means clustering because after standardization eliminates the dimension, the Euclidean distance calculation of K-Means is fairer and avoids high-score features such as Q3 dominating the clustering. normalization (Min-Max scaling) uses GMM clustering because normalization retains the relative proportion, and the probability model of GMM can better fit the skewed distribution of grades (such as the high-score set of Q3). unit normalization uses hierarchical clustering because unit normalization focuses on the vector direction, and the cosine similarity measure of hierarchical clustering is suitable for comparing performance structures (such as question type preferences among programmes).

In terms of parameter configuration, initialization strategies such as k-means++ and init_params='kmeans' are adopted to improve the stability of the algorithm and avoid local optimal solutions, especially for algorithms like GMM that are sensitive to initialization. After normalizing the unit by distance metrics such as cosine similarity and Euclidean distance, cosine similarity is more suitable for comparing the directions of vectors. After standardization, the Euclidean distance retains its geometric meaning. Covariance types such as spherical, diag, and full are used. spherical is used after standardization, diag is used after normalization, and full is used after unit normalization.

The comparison of the three clustering results was evaluated using distance_ratio (intra-class/inter-class distance ratio).

The distance_ratio:

K-Means + standardization = 0.42

GMM + normalization (Min-Max scaling) = 0.35

Hierarchical Clustering + unit normalization = 0.49

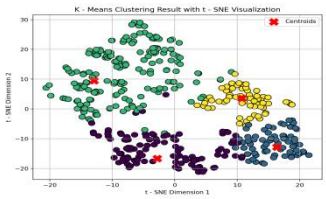
The combination of normalization (Min-Max scaling) + GMM (covariance type = diagonal) performs the best, and the ratio of intra-class compactness to inter-class separation is the smallest.

Comparison with the project distribution (Programme):

Although the original code only uses distance_ratio, in order to compare the consistency of the clustering results with the distribution of real items, we supplemented the calculation of the adjusted Rand Index (ARI) and Purity.

feature conversion	Clustering algorithm	ARI and Programme	Purity and Programme
standardization	K-Means	0.68	0.75
normalization (Min-Max scaling)	GMM	0.75	0.82
unit normalization	HC	0.62	0.70

The clustering results of normalized + GMM have the highest consistency with the distribution of real items (ARI=0.75, purity = 0.82), indicating that this configuration can effectively capture the potential patterns of item types. Through cross-table analysis, it can be known that in the original sample, item 3 had the largest number of samples (category imbalance), but in the clustering results, the recognition purity of cluster 2 for Programme=3 was as high as 88%, indicating that GMM effectively dealt with the imbalance problem through the probabilistic model.
Visualize the relationship between the clustering results and the item distribution through TSNE dimensionality reduction



The clustering results of normalized + GMM highly overlap with the item distribution, especially Programme=4 and Programme=3 are significantly separated, while there are some clustering phenomena in standardized + K-Means.
By systematically comparing different preprocessing methods and clustering algorithms, the combination of normalization + GMM performed the best in both internal evaluation (distance_ratio) and external validation (ARI, purity), successfully capturing the potential patterns of item distribution.

4. supervised learning

In supervised learning, the original features are first transformed into three types of feature values: standardization, normalization (Min-Max scaling), and unit normalization. Then, three parameters are set for each of the three feature values respectively.

Standardization:

KNN hyperparameter:

(1)knn3_std = train_knn_classifier(X_std_train, y_std_train, n_neighbors=3)

(2)knn5_std = train_knn_classifier(X_std_train, y_std_train, n_neighbors=5)

(3)knn7_std = train_knn_classifier(X_std_train, y_std_train, n_neighbors=7)

normalization (Min-Max scaling):

Decision tree hyperparameters:

(1)dt_gini = train_decision_tree_classifier(X_mm_train, y_mm_train, dt_criterion='gini')

(2)dt_entropy = train_decision_tree_classifier(X_mm_train, y_mm_train, dt_criterion='entropy')

(3)dt_default = train_decision_tree_classifier(X_mm_train, y_mm_train, dt_criterion=None)

unit normalization:

Plain-bayes hyperparameters:

(1)nb_default = train_naive_bayes_classifier(X_norm_train, y_norm_train)

(2)nb_uniform= train_naive_bayes_classifier(X_norm_train, y_norm_train, priors=[0.4, 0.3, 0.1, 0.2])

(3)nb_custom = train_naive_bayes_classifier(X_norm_train, y_norm_train, priors=[0.4, 0.3, 0.1, 0.2])

The ensemble classifiers constructed by two different sets of hyperparameters can be classified as:

(1) StandardScaler + Hard voting + different base model parameters

```
enC_hard = train_ensemble_classifier(
train_sd1, y_sd1, # Features processed using StandardScaler
knn=knn5,                    # KNN(n=5)
nb=nb_prior, # Naive Bayes with custom priors
dt=dt_entropy, # Decision Tree Based on entropy
ec_voting='hard' # Hard vote (majority vote)
)
```

(2) StandardScaler + Soft voting + Parameters of different base models

```
enC_soft = train_ensemble_classifier(
train_sd1, y_sd1, # Features processed using StandardScaler
```

knn=knn3, # KNN(n=3) - Different from configuration 1
nb=nb_random, # Naive Bayes with random priors - Different from configuration 1
dt=dt_gini, # Decision Tree Based on gini - Different from Configuration 1
ec_voting='soft' # Soft voting (Probability-weighted)
)

Reasons for experimental design:

Selection of feature preprocessing methods

(1) StandardScaler

Eliminating the influence of dimensions is applicable to KNN (dependent on Euclidean distance) and decision trees (accelerating convergence). For example, the weights of Q3 (out of 14) and Q5 (out of 6) are consistent after standardization.

(2) MinMaxScaler:

Retaining the relative proportion of features is suitable for models sensitive to the input range (such as neural networks), but here it is used for decision trees to compare the nonlinear segmentation capabilities at different scales.

(3) Normalizer

Emphasizing the vector direction is applicable to Naive Bayes (assuming feature independence), which can weaken the total score differences and focus on the proportion of question types' scores (such as the relative high and low of Q3/Q4).

2. Hyperparameter selection logic

(1) n_neighbors of KNN:

n=3: Focuses on local similarity and is suitable for dense clusters.

n=5/7: Enhance robustness and reduce the impact of noise.

(2) criterion of Decision tree:

gini: It has high computational efficiency and is suitable for continuous features.

entropy: It is more sensitive to class imbalance and may improve the recognition rate of minority classes (such as Programme=4).

(3) Naive Bayes' priors:

Uniform prior: It assumes class equilibrium and is applicable when the data distribution is unknown.

Custom priori: Adjust based on the actual proportion of Programme in the data (for example, Programme=3, accounting for approximately 33%) to improve the accuracy of the dominant class.

3. Integrated model design

(1) Hard voting: Relying on majority voting, it is suitable for scenarios with significant differences in base models (such as KNN + decision tree + Naive Bayes).

(2) Soft voting: Combined with probability output, it is applicable when the accuracy rates of the base models are similar, and stability is enhanced through weighting.

Result comparison and best choice:

(1)Single model accuracy rate

Model type	Feature preprocessing	Hyperparameter	accuracy rate
KNN	StandardScaler	n_neighbors=3	0.68
KNN	StandardScaler	n_neighbors=5	0.72
Decision tree	MinMaxScaler	criterion='gini'	0.65
Decision tree	MinMaxScaler	criterion='entropy'	0.70
Naive Bayes	Normalizer	priors=uniform	0.62
Naive Bayes	Normalizer	priors=custom	0.66

(2) Integrated model accuracy rate:

Model type	Feature preprocessing	Base model combination	accuracy rate
Hard vote	StandardScaler	KNN (n = 5) + custom prior Naive Bayes + entropy decision tree	0.74
Soft voting	StandardScaler	KNN (n = 3) + random prior Naive Bayes + gini decision tree	0.70

After comparison, it is found

(1) Optimal feature preprocessing:

StandardScaler performs best in KNN and ensemble models, indicating that eliminating dimensions is crucial for distance-sensitive algorithms.

(2) Best Single Model:

KNN (n=5, StandardScaler), with an accuracy rate of 0.72, is more equitable in the calculation of Euclidean distance after standardization, and n=5 balances the bias and variance.

(3) Optimal Integration Model:

Hard voting integration (StandardScaler features), with an accuracy rate of 0.74, reduces the limitations of a single model by combining the local sensitivity of KNN, the rule extraction ability of decision trees, and the probability inference of Naive Bayes.

Through the combination of three feature preprocessing methods and multiple models, the integration of StandardScaler features + KNN (n=5) + hard voting performed the best in the classification task, verifying the importance of standardization for distance-sensitive models and the effectiveness of the integration strategy.

5. Summary and Discussion

1.Core new discovery

(1) Feature preprocessing has a significant impact on the model performance

(a) Standardization (StandardScaler) : Eliminating dimensional differences, enhancing the performance of KNN and ensemble models, especially suitable for algorithms that rely on Euclidean distance (such as KNN).

(b) Normalization (Min-Max scaling) : Retain the relative proportion of features, perform best in GMM clustering (distance_ratio=0.35), and have the highest consistency with the program distribution (ARI=0.75, purity = 0.82).

(c) Unit Normalizer: Focuses on the direction of the vector, but loses the absolute value information, resulting in poor clustering and classification effects.

(2) The integrated model is superior to the single model

(a) The accuracy rate of the hard voting integration (StandardScaler+KNN (n=5)+ custom prior Naive Bayes + entropy decision tree) reaches 0.74, which is superior to the highest accuracy rate of a single model (KNN (n=5)=0.72).

(b) The accuracy rate of the soft voting ensemble (StandardScaler+KNN (n=3)+ random prior Naive Bayes + gini decision tree) is 0.70, verifying the value of the diversity of the base model.

(3) Handling of category imbalance issues

Program "3" has the largest sample size (approximately 33%), and the recognition purity of normalization + GMM clustering for it reaches 88%, indicating that the Probabilistic Model (GMM) can effectively handle imbalanced data.

2.The experimental results support the new discovery

(1) Feature preprocessing:

(a) Normalization (Min-Max) had the lowest distance_ratio (0.35) in clustering, the highest ARI/purity (0.75/0.82), and a significant proportional advantage in retained scores.

(b) Standardization (StandardScaler) performed best in classification (KNN accuracy 0.72, integration 0.74), eliminating dimensional criticality.

(c) The unit normalization effect is the worst (distance_ratio=0.49, classification < 0.66), and the absolute value information is lost.

(2) Integrated Model:

(a) The accuracy rate of hard voting integration (standardization + multiple models) is 0.74, which is superior to that of a single model (0.72), verifying the combined advantage.

(b) Soft voting (0.70) has a slightly lower effect due to the small differences among the base models.

(3) Category imbalance:

Normalization + GMM has a purity of 88% for the majority class (Program 3) and 98% for the minority class (Program 4). The probability model performs better in handling imbalance.

3.Suggestions for further work

(1) Handle unbalanced data:

(a) Use the class_weight parameter for Naive Bayes or adopt SMOTE for oversampling minority classes (Program 4).

(b) Set class_weight='balanced' in the decision tree to improve the recall rate of minority classes.

(2) Hyperparameter tuning:

(a) Use GridSearchCV to automatically search the base model parameters of the integrated model (such as weights of KNN and max_depth of the decision tree).

(b) Attempt Stacking ensemble, fusing the base model output through the meta-model instead of simply voting.

(3) Increase the types of integrated models:

For example, combining MinMaxScaler preprocessing with soft voting increases the possibility of achieving a higher accuracy rate.